EE3032 Introduction to VLSI Design—Homework 2

1. **Multi-V_t Design** (3 points)
   Fig. 1(a) shows a combinational logic in which each logic gate has 1 unit delay if low V_t transistors are used to realize the logic gates. If we want to reduce the subthreshold conduction power of the circuit using dual-V_t process (i.e., you have high-V_t and low-V_t transistors) without sacrificing the performance of the circuit, please draw the transistor-level schematic of the combinational circuit with dual-V_t transistors using the symbols shown in Fig. 1(b).

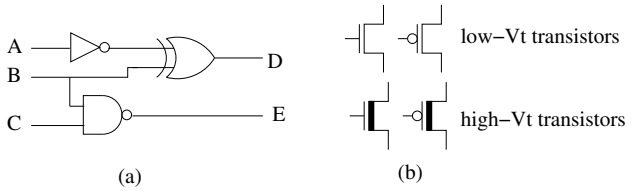

Figure 1: (a) A simple CMOS logic (b) Symbols of transistors

2. **CMOS Design** (3 points)
   Design a CMOS circuit to implement the truth table shown in Table 1. This circuit has 4 data inputs (a_0–a_3), 4 control inputs (c_0–c_3), and 4 data outputs (z_0–z_3).

Table 1: Truth table of a CMOS circuit.

c_3	c_2	c_1	c_0	z_3	z_2	z_1	z_0
0	1	1	1	a_3	a_2	a_1	a_0
1	0	1	1	a_2	a_1	a_0	a_3
1	1	0	1	a_1	a_0	a_3	a_2
1	1	1	0	a_0	a_3	a_2	a_1

3. **Logic Gate** (4 points)
   Fig. 2 shows a CMOS logic gate. If the size of NMOS and PMOS transistors is W/L, the corresponding turned-on resistance is R_n and R_p, respectively, regardless of the transistor is operated at linear or saturation regions. Also, $R_p = 2R_n$, where $\mu_n = 2\mu_p$ is assumed. Complete the following questions.

   (a) What is the Boolean function realized by the logic gate? (1 points)

   (b) If the highest voltage of logic 0 is 0.1Vdd, determine the sizes of the transistors such that the logic gate can operate correctly. (3 points)

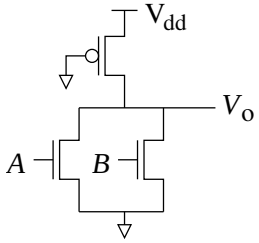

Figure 2: CMOS logic gate